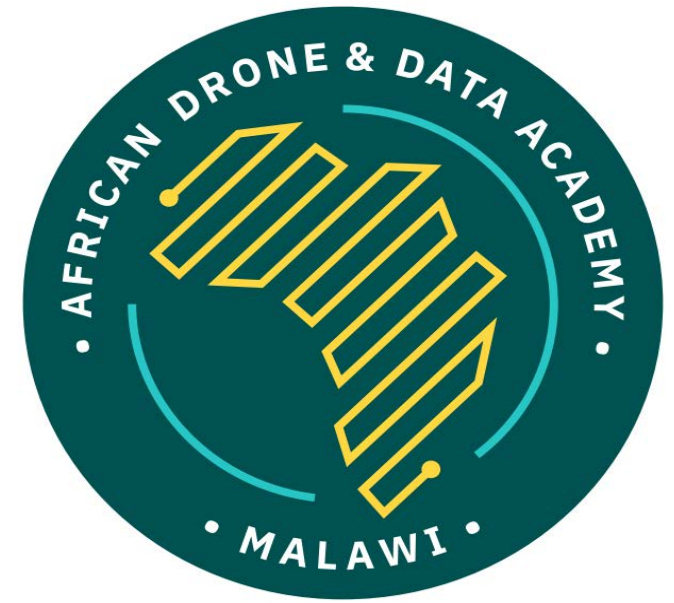




TOP Level 1

Day 4 Version A





TOP Level 1

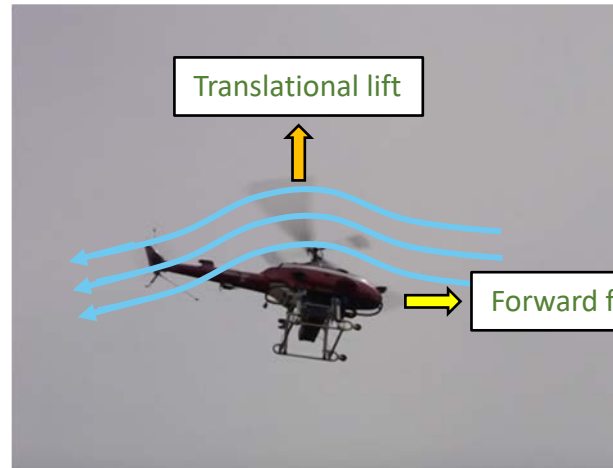
Principles of Flight Part 3



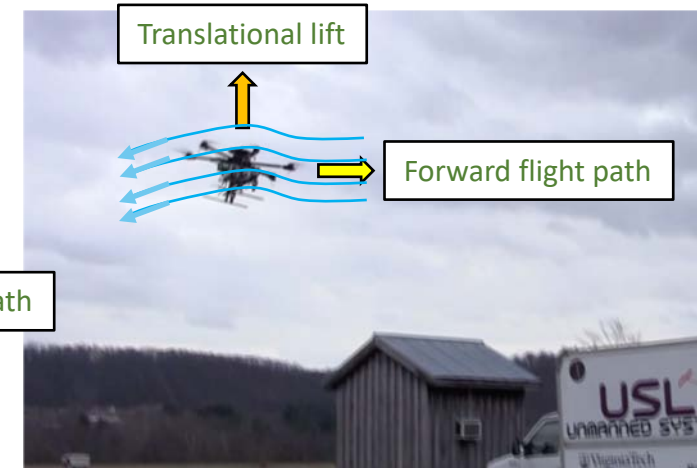
Flight performance of VTOL aircraft

- VTOL aircraft have an advantage that they can hover, however we have seen that they consume much more power than a fixed wing aircraft
- VTOL are more efficient when flying forward as opposed to hovering
 - This is called “translational lift”
- Translational lift is **additional lift** that a VTOL aircraft produces when it **flies forward**
- The downwash created by the rotor(s) act similar to wing with camber that turns the relative air downward

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. **Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



RMAX helicopter



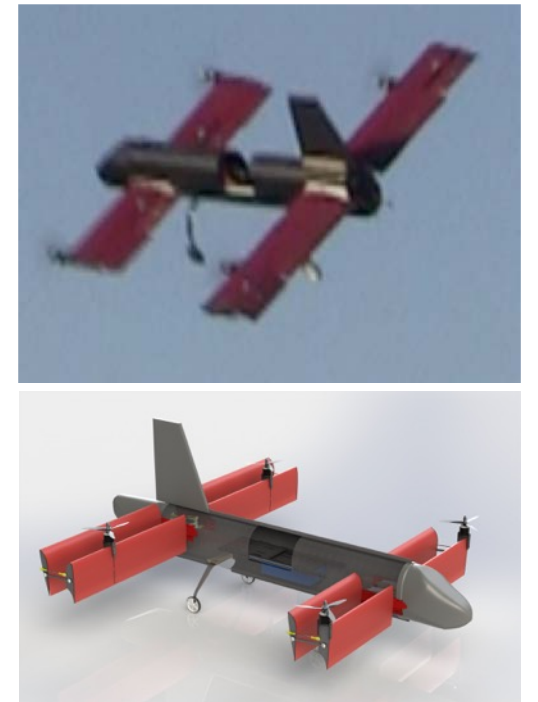
Hexacopter

Power consumption may decrease 20% for certain multi-rotor aircraft that are flown with forward speed

VTOL flight control

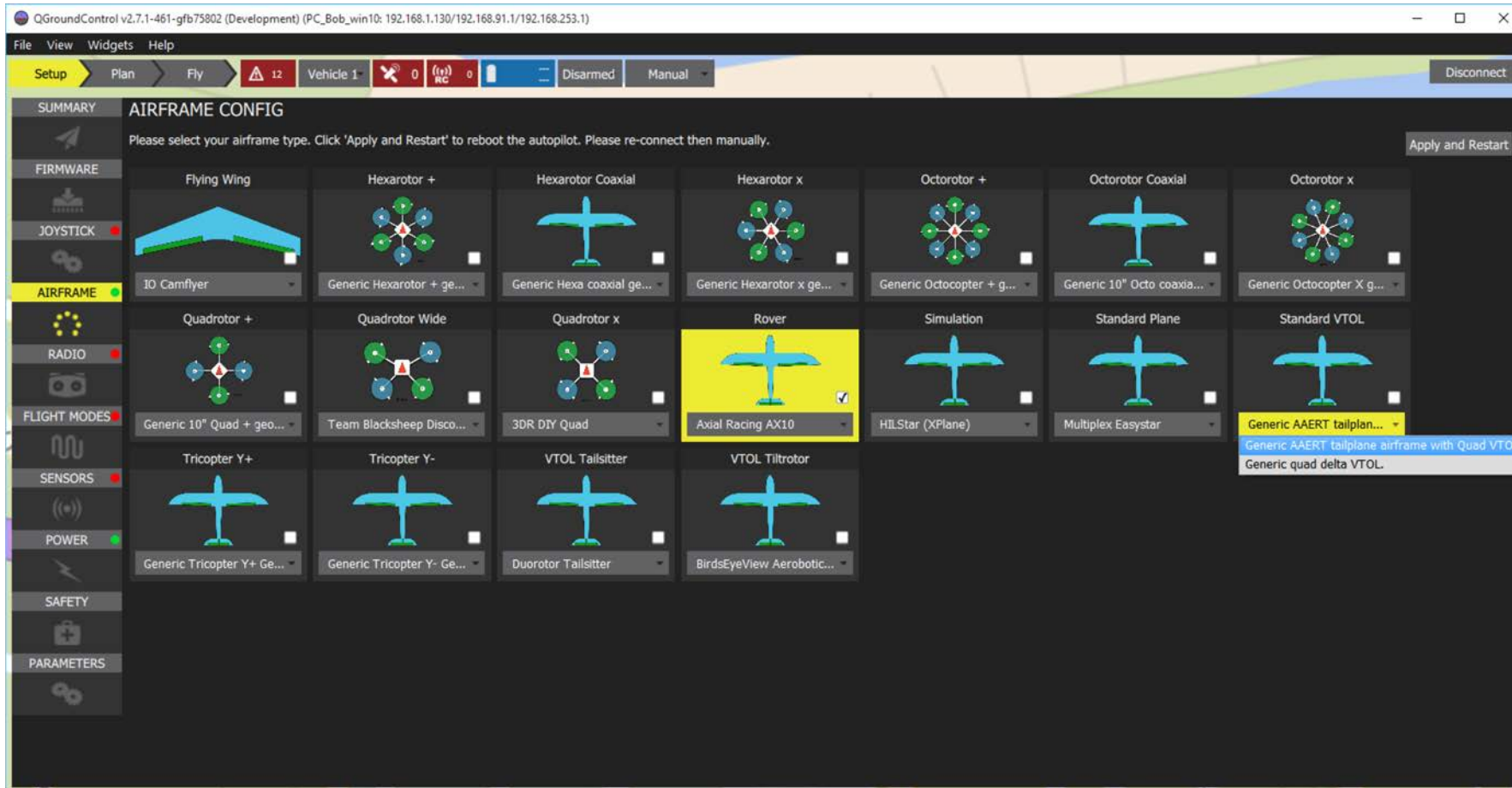
- One of the reasons that multirotor aircraft are so popular is that they are relatively easy to control using modern electronic systems
 - High processor speeds on relatively small form factors allow high bandwidth control applied to a system of several motors
 - The **Pixhawk 4 mini**, which we use, is an open-source platform for developing novel flight control programs
 - There are many aircraft configurations already available to “plug into” the flight controller
 - If a new flight control scheme is needed, the code can be written to accomplish the flight control task

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



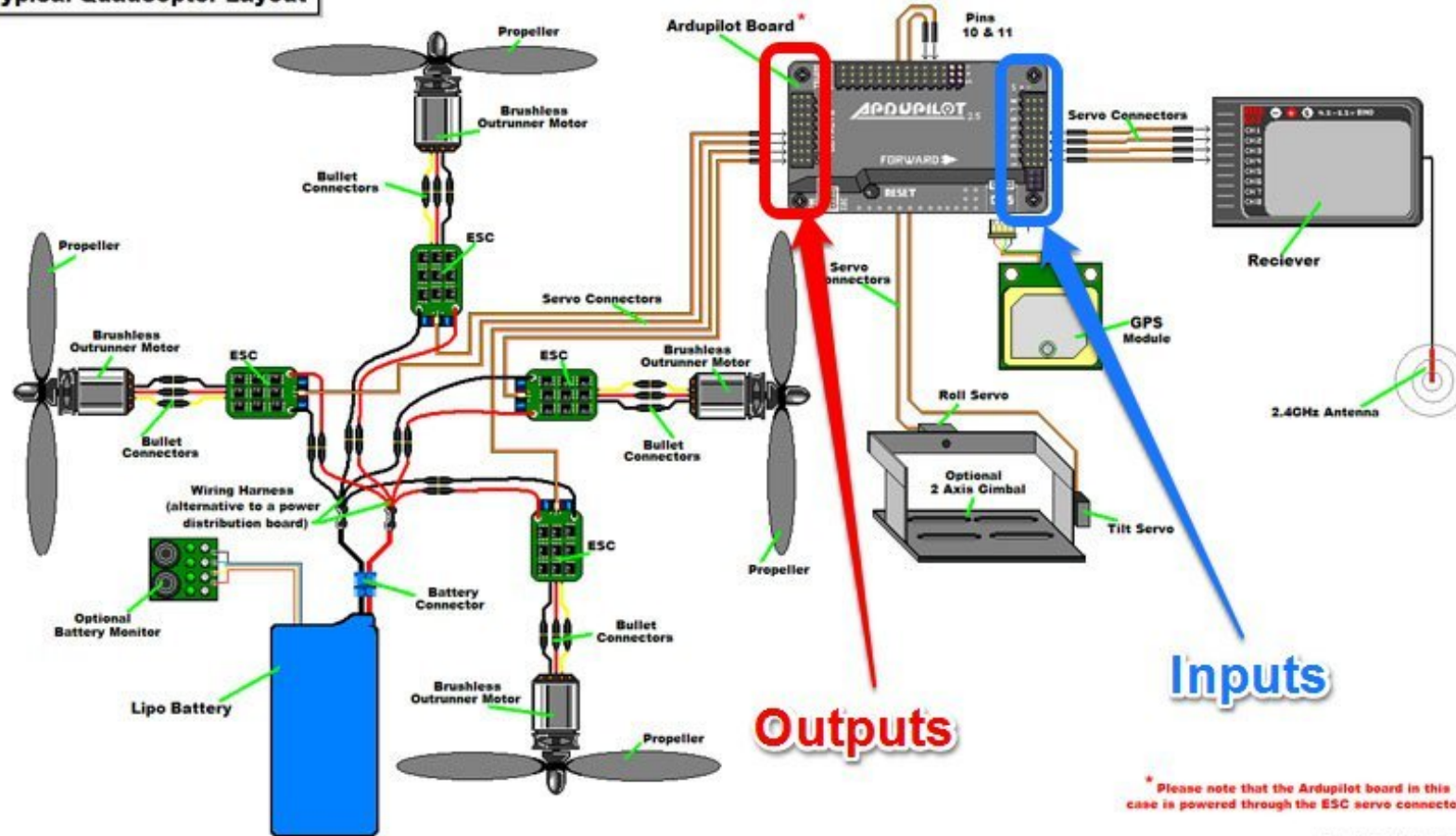
VTOL flight control: available configurations to select in QGroundControl

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



VTOL flight control - the layout of hardware and signal routing

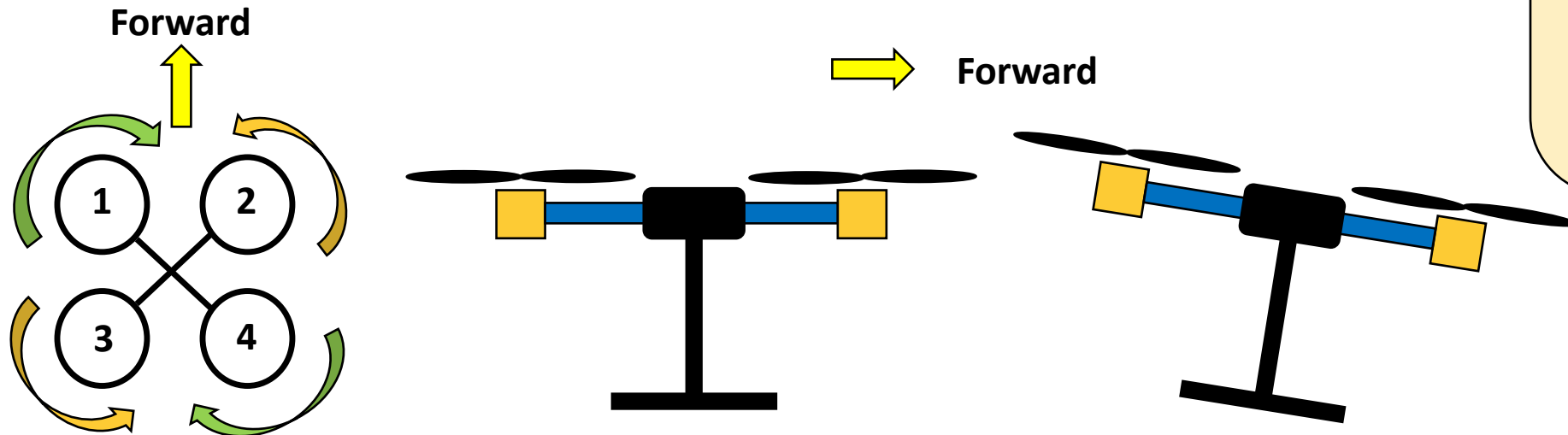
Typical Quadcopter Layout



11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

VTOL control: how flight control functions on a quadcopter

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



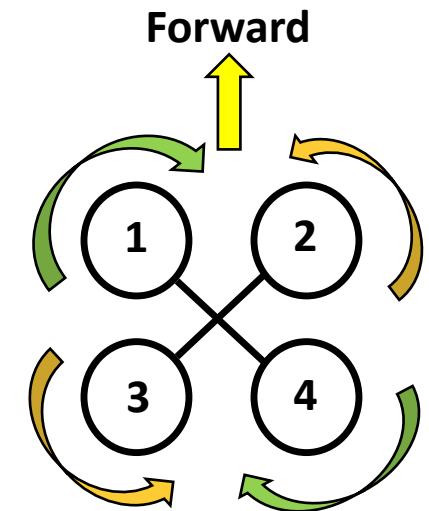
- The motors on the quadcopter turn in alternating directions to cancel the torque of the system
- Changes in RPM of rotor pairs results in pitch, roll and yaw control of the aircraft
- For instance, to pitch the aircraft forward, rotor speeds for 1 and 2 are decreased and 3 and 4 are increased, while the net thrust is not changed

VTOL control: how flight control functions on a quadcopter

- Likewise, changing the RPM of different combinations of rotors will affectively maneuver the aircraft in all axes

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12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

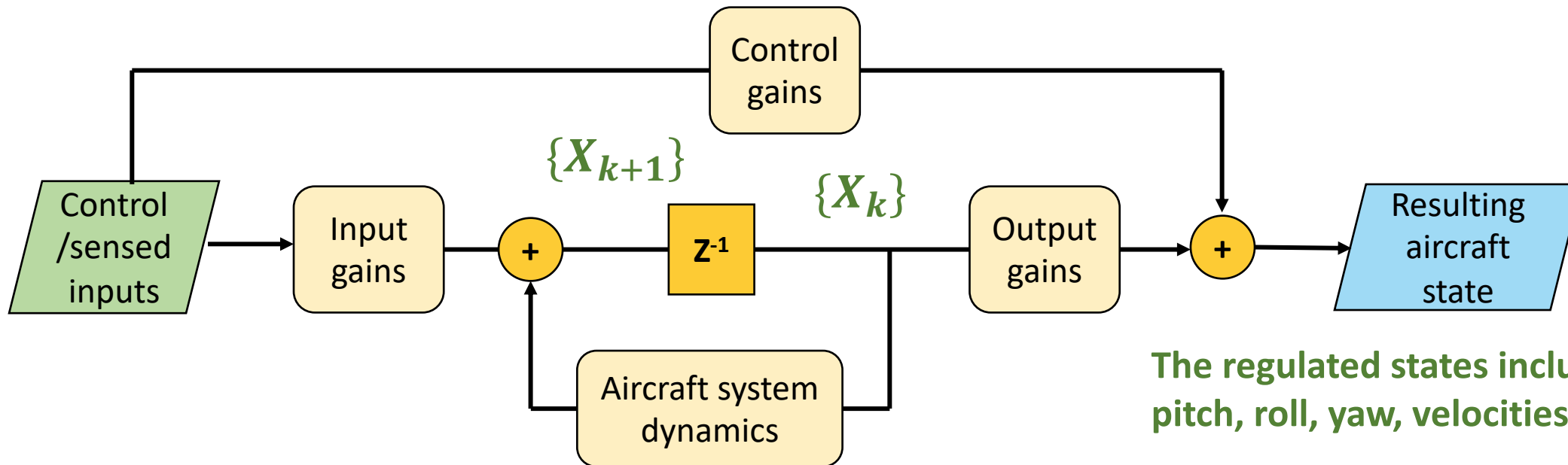
Maneuver	Rotor speed change
Pitch forward	Decrease 1,2 and increase 3,4
Roll right	Decrease 2,4 and increase 1, 3
Yaw left	Increase 1, 4 and decrease 2, 3



VTOL control: how flight control functions on a quadcopter

- The Pixhawk 4 Mini contains a full inertial measurement unit (IMU) to sense the state of the aircraft and provide feedback to regulate the RPM of the motors

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



The regulated states include aircraft pitch, roll, yaw, velocities and positions

VTOL control: how flight control functions on a quadcopter

- The sensed aircraft states by the flight controller require sensors and electronics that convert the analog values to digital values
 - Magnetometer senses attitude (pitch, roll and yaw)
 - MEMS gyro senses pitch, roll and yaw rates
 - Accelerometers sense accelerations $\ddot{x}$, $\ddot{y}$, $\ddot{z}$
 - A MEMS pressure sensor senses barometric pressure
 - A digital airspeed sensor (on fixed wing) senses airspeed
 - The GPS receiver senses position

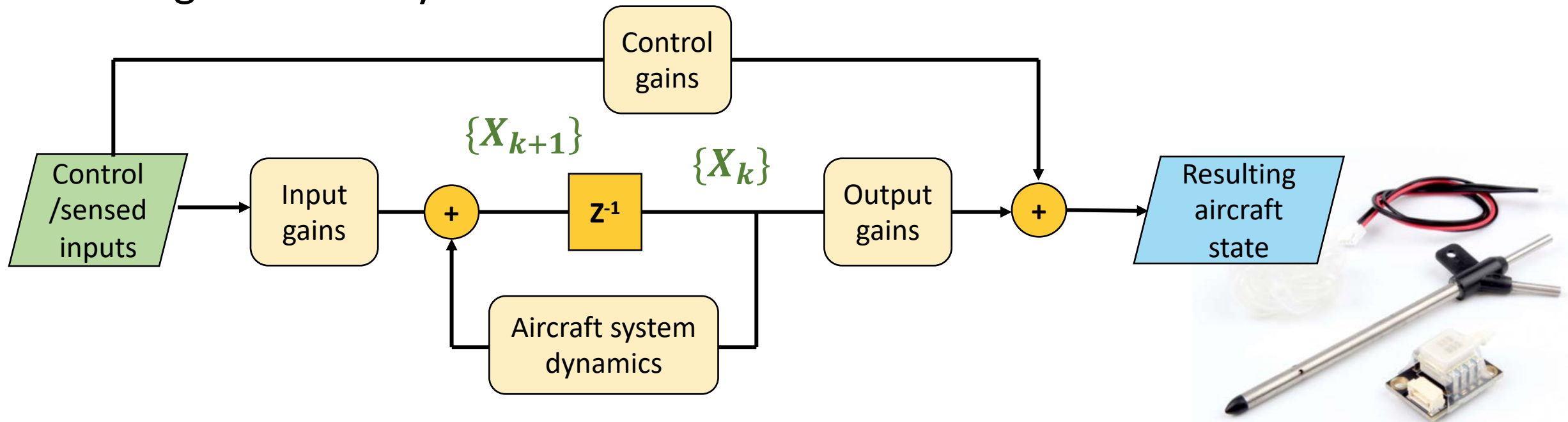
11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



Fixed wing flight control operates similar to VTOL using state feedback

- The addition of airspeed sensing on a fixed wing aircraft provides a valuable velocity input to the flight control system

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. **Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



All flight control systems will be compromised if a sensor fails or is providing erroneous readings

- EcoSoar at one time had problems with the airspeed sensor
 - This could have been poor electrical connections or incorrect sensed air data readings
- A flight controller presented with incorrect low airspeed readings will command the aircraft to **pitch down** to **speed up**
 - This has resulted in crashes in the past

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2





TOP Level 1

Flight Performance



Flight performance, planning and loading

- All flights must comply with the laws of physics!
- The following rules should be followed to launch a successful mission:
 1. **Loading** of the aircraft should not exceed the design limits as the performance is directly related to the weight carried
 2. **In mission planning**, it is important to validate the aircraft will have **adequate range** and endurance to complete the mission
 3. **Environmental conditions** directly affect aircraft performance and must be evaluated carefully

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



VTOL aircraft loading

- Weight has a direct impact on aircraft performance, and keeping the payload loading of the aircraft within its design limits is critical to safe flight performance
- Payload **weight that changes** during flight, such as **sprayer payloads**, need to consider both aircraft weight and balance during the mission
- Balance of the aircraft is just as important as the weight - **the flight control system assumes that the center of gravity is near the center of the airframe**
 - Any exceedances of center of gravity position can result in an inability of the flight controller to maintain stable flight

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Aircraft loading considerations: density altitude

- Density altitude is a strong factor in aircraft performance and is always determined and used in the flight planning process
- The density altitude is an effective altitude for a “standard day” ($T = 20^{\circ}\text{C}$, $P = 1.01\text{E}^5 \text{ N/m}^2$) and the actual atmospheric conditions which include altitude, humidity and temperature
- Density altitude is calculated using the ideal gas equation to find a corrected density due to the measured temperature: $\rho = \frac{p}{RT}$

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Aircraft loading considerations: density altitude

- When flying in hot conditions (above standard temperature), the density altitude can be much higher than the actual altitude
- Since lift is a direct function of air density, a **decrease in density** will result in a **decrease in lift**
 - This will reduce the payload you are able to carry on the aircraft
- In addition to reducing lift that is generated by the wings or propeller, internal combustion engines will also suffer a power loss as the air density decreases

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Aircraft loading considerations: density altitude

- Compute the density altitude in Lilongwe on a 35°C day:

Altitude = **1100m**

Standard day pressure = $p = 8.879\text{E}^4 \text{ N/m}^2$

$$\rho = \frac{p}{RT} = \frac{8.879 \times 10^4}{287 (308.15\text{K})} = 1.004 \text{ kg/m}^3$$

- Looking up this density in the standard day atmospheric charts shows that the corresponding altitude would be **$h = 2000\text{m}$**
- Therefore, the aircraft thinks it is flying at a much higher altitude due to this much higher temperature

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



The effect of density altitude on speed and performance

- Airspeed sensing on aircraft use a ***pitot probe*** that measures **dynamic pressure**
- **Dynamic pressure** is the pressure resulting from Bernoulli's principle that indicates a **reduction in flow velocity** results in **higher pressure**:
$$p_{dynamic} = \frac{1}{2} \rho V^2$$
- The velocity is **reduced all the way to zero** with a pitot probe, which means that the measured pressure is **directly proportional** to the air density and quadratically related to the **speed of the aircraft (velocity²)**

11. RPA operations
12. **Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



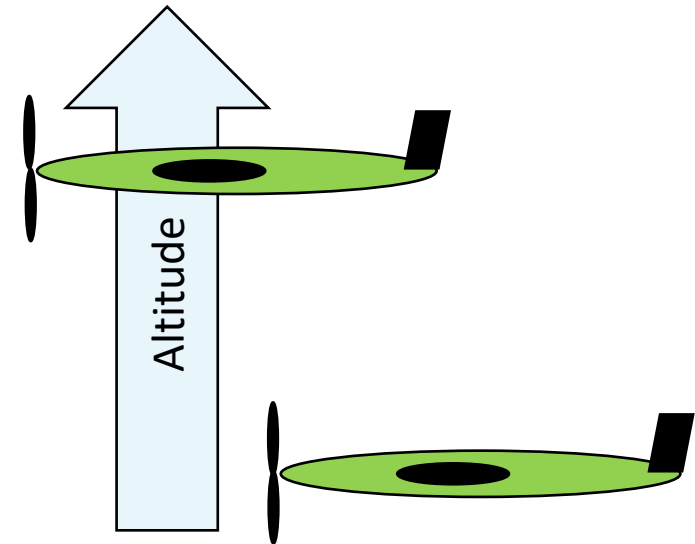
The aircraft only knows dynamic pressure, and so actual (true) airspeed will change with altitude (air density)

- **Indicated airspeed** is what is directly read from the pitot probe using a sea level reference density altitude:

$$V = \sqrt{2p_{\text{dynamic}} / \rho_{\text{sea level}}}$$

- **True airspeed** is the actual airspeed which is corrected using the actual air density
- If the **thrust remains the same** as the airplane **climbs in altitude**, the aircraft **will actually fly faster** because of the **lower air density**
 - However, to keep the same thrust, the speed of the propeller must increase because of the lower density

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

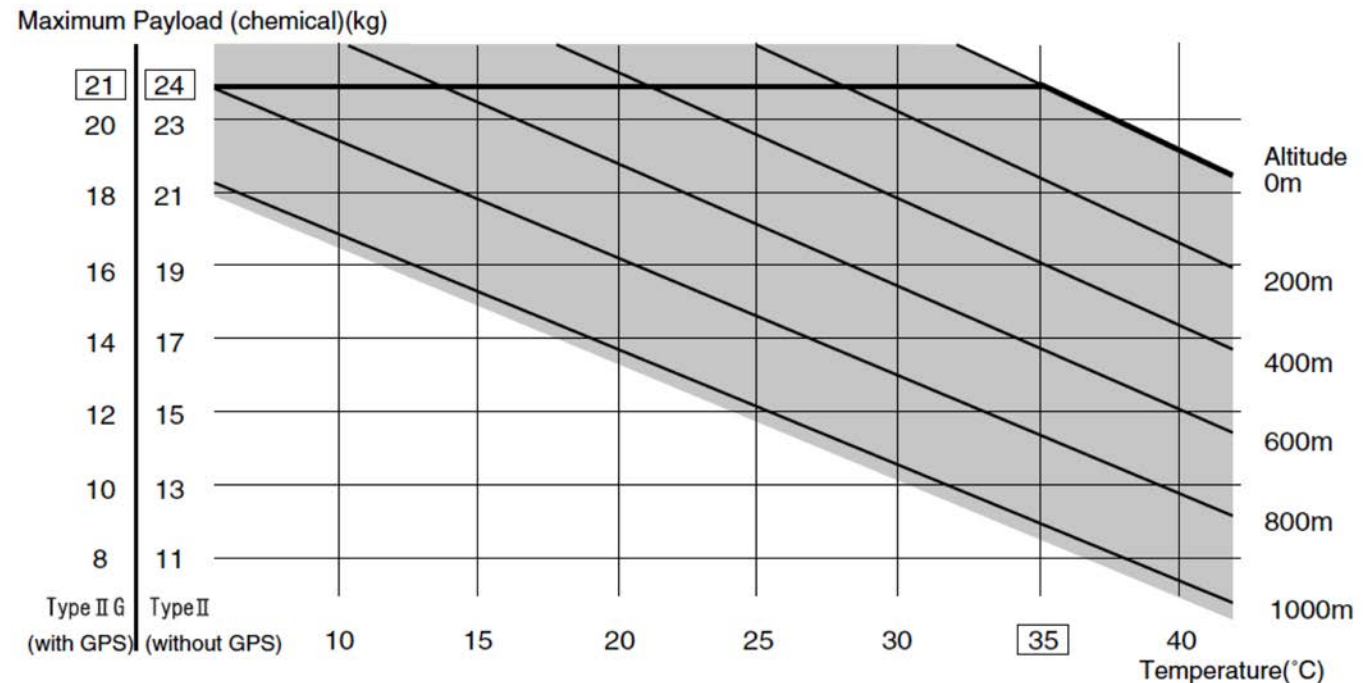


Because the air density is **lower** at higher altitudes, the aircraft will fly **faster** if the thrust remains constant

Aircraft loading considerations: density altitude

- Some aircraft manufacturers will specify payload loading as a function of altitude and temperature
- The chart shown for the Yamaha RMAX helicopter is typical of a density altitude loading chart that will guide the operator to safe loading of the aircraft
- If a chart does not exist, then the reduction (fraction) of lift can be approximated by using a ratio of flight altitude density to sea level density

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Aircraft loading considerations: density altitude

- Propeller thrust, like lift and drag, can be described by a non-dimensional thrust coefficient: $T = C_T \rho n^2 d^4$

Here, C_T is the thrust coefficient

ρ is the air density, n is the rotational speed in cycles/sec, and d is the propeller diameter

- Since the thrust is directly proportional to air density, it is obvious that if the **air density decreases by 10%**, then the **thrust or lifting capacity of the aircraft will also decrease by 10%**

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Aircraft loading considerations: aircraft balance

- In addition to a need for the aircraft weight to be within an acceptable range, the **position** of the **center of gravity (CG)** of the aircraft is important
- The CG is important for both VTOL and fixed wing aircraft since aerodynamically the aircraft is designed to be balanced about a fixed location
 - A CG that is not at the design point or falls within an acceptable range will cause the aircraft to be unstable or uncontrollable, resulting in a crash

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Aircraft loading considerations: aircraft balance

- A balance test is always performed prior to flight to verify visually that the CG is in the right place.
- For multi-rotor aircraft, this test is more of a qualitative test performed by loosely holding onto opposite arms of the aircraft to see what angle the aircraft moves to
 - It should not be angled more than a few degrees from vertical
- For fixed wing aircraft, the balance point is 25% of the distance from the effective leading edge of the wing
 - The effective leading edge is computed geometrically by accounting for wing sweep

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Validate that the aircraft has necessary range and endurance

- As part of the flight planning activity, the aircraft range and time aloft should be estimated so that it is reasonably assured the aircraft will be able to complete its mission
- Time aloft estimate: based on the mission plan estimate of completion time for the mission
 - Note that the flight plan may not account for takeoff and landing that should be factored into an overall endurance estimate
- An accurate aircraft endurance estimate is based on past experience
 - If the temperature is significantly different, then that will either add or subtract to the endurance estimate

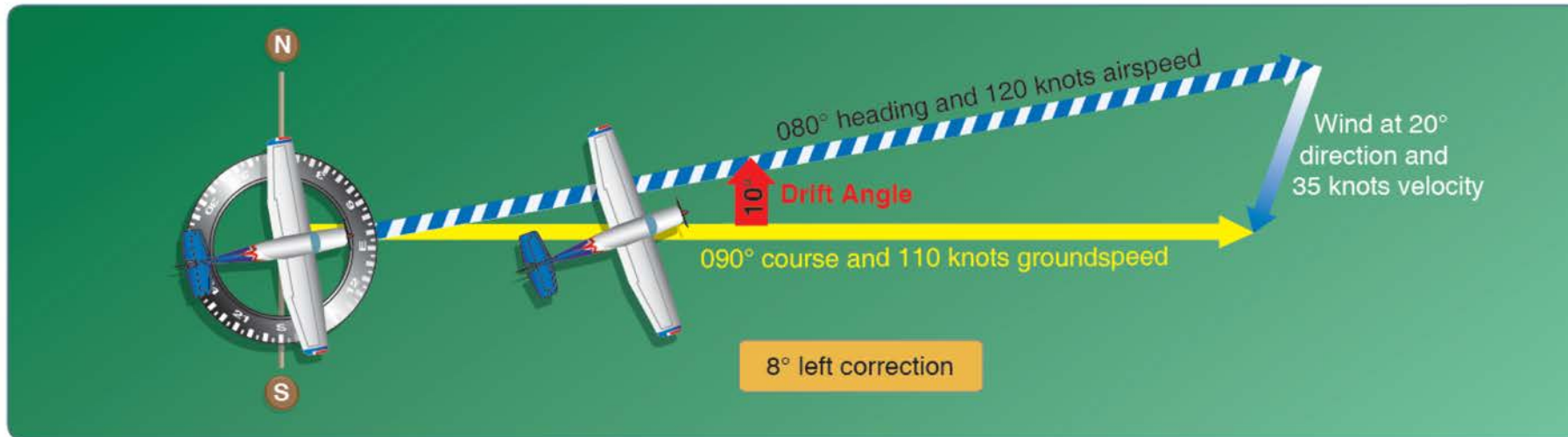
11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Validate that the aircraft has necessary range and endurance

- With fixed wing aircraft that are traveling at a much higher speed at higher altitude, it is advisable to consider the winds aloft when determining how long the flight will take

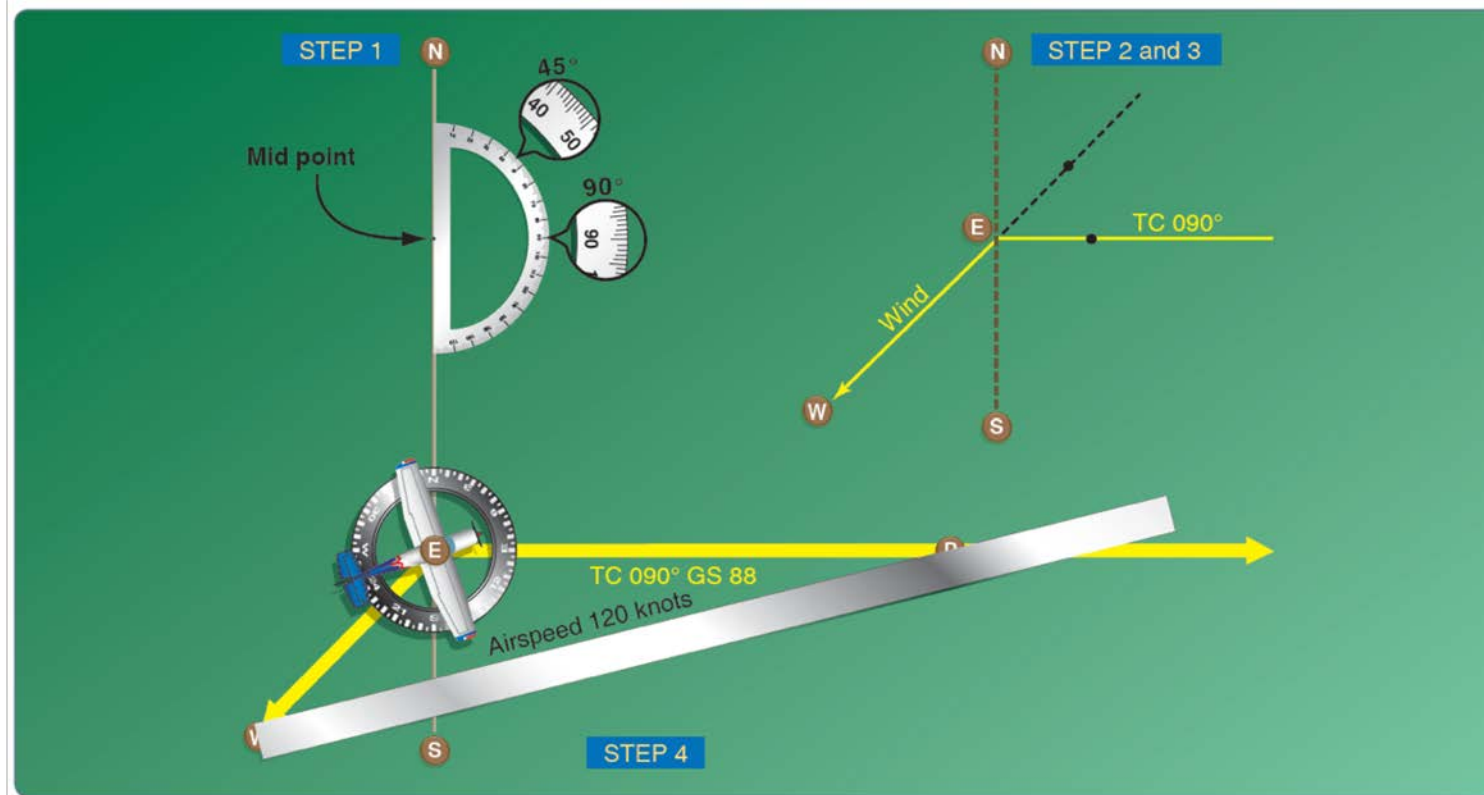
11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



An effective groundspeed is determined once a wind triangle has been drawn to find the drift component and the compensating flight heading to maintain course

Computing the crosstrack correction and
groundspeed will provide accurate mission
times to determine feasibility

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



To determine aircraft groundspeed, plot the wind vector and the aircraft vector to find the flight heading + groundspeed over the desired track. Check endurance with battery capacity

Consider environmental conditions when planning a mission

- The operating environment has a strong influence on the performance of RPA
 - Weather considerations need to be addressed in flight planning to reveal any adverse conditions that may occur before or during a flight activity
 - Hot, dry weather can cause dust devils to occur which are very hazardous to RPA operations
 - A dust devil is a small tornado, capable of causing the flight control system to be overwhelmed and leading to a crash
 - Convective weather activity can indicate wind shears and strong frontal boundaries with rapidly increasing wind

11. RPA operations
- 12. Flight performance**
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

